

Mathematics II

UNIT-I

Integral calculus: Bernouli's Formula, Reduction Formula $\sin^n x$, $\cos^n x$, $\sin x \cos x$ – Simple problems.

RTB(1): Chapter 2: Sections: 2.5.5, 2.6.2

UNIT-II

Fourier Series: Fourier series for functions $(0, 2\pi)$, $(-\pi, \pi)$

RTB(1): Chapter 3: Sections: 3.1.1, 3.1.2

UNIT-III

Differential Equations: Ordinary Differential Equations: second order non-homogeneous differential equations with constant coefficients of the form $ay'' + by' + cy = X$ where X is of the form $e^{ax} \cos \beta x$ and $e^{ax} \sin \beta x$ - Related problems only.

Partial Differential Equations: Formation, complete integrals and general integrals, four standard types and solving Lagrange's linear equation $Pp + Qq = R$.

RTB(1): Chapter 4: Sections: 4.3.1, Chapter 5: Sections: 5.1.1, 5.4.1

UNIT-IV

Laplace Transforms: Laplace transformations of standard functions and simple properties, inverse Laplace transforms.

RTB(1): Chapter 6: Sections: 6.1.1, 6.1.2, 6.2.1

UNIT-V

Vector Differentiation: Introduction, Scalar point functions, Vector

Point functions, vector differential operator Gradient, Divergence, Curl, Solenoidal, irrotational, identities.

RTB(1): Chapter 7: Sections: 7.1.1 to 7.1.3, 7.2.1 to 7.2.5, 7.3.1 to 7.3.4